OC Core 1.3.1 Critique Closure Companion EN

Document boundary

- This critique-closure companion is the source-audit and objection-handling route for OC Core 1.3.1.
- It is built from the actual source audit, theorem roadmap, and reviewer-navigation corpus.
- The 1.3.1 critique appendix records bounded closure, clarification, or demotion outcomes only.

Contents

1 Reader Contract and Navigation Guide

Core 1.3 is the flagship reader-facing monograph, but readers arrive with different tasks. This guide is therefore a route map rather than a second introduction. Its job is to show where each of the five reader roles should begin: Reviewer route, Formal reader route, Domain reader route, Broad scientific reader route, and Practical user route. External collaborators who need deployment guidance should use the Practical user route.

The dedicated Figure Atlas in Appendix **??** and the Technical Derivation Atlas in Appendix **??** are intentionally outside the default continuous reading route. Their material belongs to the appendix inspection layer; the formal-reader route below consults the Technical Derivation Atlas only when local proof support is needed.

1.1 The five reading routes

The reviewer-navigation appendix uses slightly narrower review labels: "rapid critical reviewer" corresponds to the Reviewer route, "formal mathematical reviewer" to the Formal reader route, and "generalist scientific reviewer" to the Broad scientific reader route.

Reviewer route. Read the front matter, this guide, Sections **??**, 2–4, Section **??**, and then Appendices A–B. That route exposes the publication boundary, theorem status, load-bearing theorem spine, reviewer objections, and source witnesses as quickly as possible.

Formal reader route. Use Sections **??**–**??** for orientation only. Then read Sections **??**–**??**, Section **??**, and Sections 2–**??** in order. Use Appendices **??**–**??**, Appendix **??**, and the Technical Derivation Atlas in Appendix **??** when local proof support is needed. This is the compact route that keeps exact doctrine, theorem scope, and proof machinery visible at once.

Domain reader route. Read Section **??** and Sections **??**–**??**, then move into Appendices **??**, **??**, **??**, **??**, **??**, and **??**, with Appendix **??** added only when experiments, predictions, or falsifiability catalogues need inspection. That path exposes bounded domain embedding, worked interpretation, empirical promotion rules, replay surfaces, benchmark cases, execution board, practical use, and comparison evidence without forcing the reader to traverse every technical-derivation section first.

Broad scientific reader route. Read Sections **??**–**??**, Sections **??**–**??**, Section **??**, and Sections 2–**??** straight through as a continuous graduate-level monograph. Use Appendix **??** for notation when needed and defer Appendices **??**–**??** until a concrete figure, benchmark table, or support dossier question arises. This is the recommended route for a broad scientific reader.

Practical user route. Read the front matter, this guide, Section 2, Section **??**, and Appendices A, **??**, and **??** first; add Appendix **??** only when the task requires experiments, predictions, or falsifiability catalogues. Return to Sections **??**–**??** only when a practical use case needs theorem, replay, or falsifier support to be inspected in detail. This route is for external collaborators who need deployment guidance: what can be predicted, which decisions can be supported, which procedures can be run, and how the result compares with alternatives.

1.2 Route legend

The five routes above are the working navigation system:

- **Reviewer route**: shortest path to claim boundary, proof spine, objections, and source witnesses.
- **Formal reader route**: compact path through doctrine, theorem scope, notation, and proof machinery.
- **Domain reader route**: bounded path from domain embedding to replay, benchmarks, execution, use, and comparison evidence.
- **Broad scientific reader route**: continuous graduate-level reading path through the main argument before atlas inspection.
- **Practical user route**: shortest path to usable tasks, required inputs, outputs, comparison boundaries, and deployment constraints.

1.3 External interaction crosswalk

The crosswalk maps common external needs to the shortest route bundle that can answer them while keeping the supporting audit trail visible.

External task	Primary route	Support route if verification is needed
Evaluate the release for review	This guide; Sections ??, 2–??; Appendices A–B	Appendix ?? when proof-obligation details are needed
Audit the bounded theorem core	Section ??, Section 3, and Section 4	Appendix ??, Appendix ??, Appendix B, and Appendix ??
Check the empirical promotion bar	Sections ??–??	Appendix ??, Appendix ??, Appendix ??, Appendix ??, Appendix ??, and Appendix ??
Cite the unified synthesis	Section ??	Appendix C, Appendix ??, and the K-level parameter tables in Appendix ??
Use or compare the model	Section ??	First inspect Appendix ??; use Appendix ?? only for experiments, predictions, or falsifiability catalogues; then use Sections ??–?? only when proof or replay support is needed
Verify publication boundaries	Section 2	Appendix A and Appendix ??

Table 1: Stable reading routes for the most common external interactions. The purpose is to keep external use concise while preserving the audit trail behind each cited claim.

1.4 What this guide is not

This guide does not restate the theory, replace the theorem roadmap, or turn the flagship monograph into a status summary. It simply tells the reader where each task belongs. Section 4 covers proof navigation. Section ?? covers the unified synthesis. Section ?? covers practical consequences and comparative analysis. The appendices keep the inspection burden inside the same volume without displacing the main reading spine.

2 Source Witnessing, Theorem Fate, and Publication Boundary

This chapter fixes the publication boundary of Core 1.3. Its task is to state which scientific authority is canonical, how source witnessing is enforced, how theorem fate is separated across outward artifacts, and how later channels must remain aligned with the source corpus, SPOT, and flagship monograph. It does not re-argue the formal doctrine or theorem spine; those burdens belong to the surrounding chapters and appendices.

2.1 Canonical scientific authority

The canonical scientific authority of the release is the SPOT projection `releases/oc_core_1_3/editorial/OC_CORE_1_3_SCIENCE_SPOT_latest.json` (surface id `OC_CORE_1_3_SCIENCE_SPOT`; current SHA recorded in its `metadata.repo_sha` field). The source-owned science corpus at `releases/oc_core_1_3/editorial/science_sources/` is the authoring and witness locus from which that SPOT is projected. Historical source witnesses, including earlier Core 1.2/Core 1.3 repair baselines, remain provenance anchors; they are not the current projection authority. The present master monograph is the flagship reader-facing projection and compiled reference volume derived from those records. It keeps the orientation ladder, formal doctrine, theorem route, empirical discipline, synthesis, practical utility, and audit layers visible together, but it does not displace the SPOT as the final authority on claim scope, promotion state, or proof boundary.

This rule matters because foundational programmes accumulate several different textual layers over time: source-native definitions and theorems, explanatory prose, repair notes, channel-specific summaries, and later operational packets. Core 1.3 treats those layers as distinguishable, not interchangeable.

2.2 Source witnessing

Core 1.3 is source-first in the strict sense that the flagship claims used in public release must remain bound to public source witnesses rather than to late summary prose alone. The point is not archival fetishism. The point is to stop drift between five things that often blur inside large research programmes:

- theorem-bearing statements,
- structural and interpretive explanation,
- repair and release notes,
- condensed public packets,
- and later operational projections.

When that boundary blurs, a summary sentence can begin to masquerade as the theorem itself, or a release dashboard can begin to replace the actual scientific record. Core 1.3 blocks that failure by keeping source witnessing bound to the source-owned science corpus and the canonical SPOT projection rather than relegating it to hidden control files.

2.3 Theorem fate and scope discipline

The present release also separates theorem fate explicitly. Not every inherited row serves the same role in every outward artifact.

- `THEOREM_NATIVE` rows may carry positive theorem-native claims when their proof and dependency route remains exact.
- `BRIDGE_ONLY` rows may remain scientifically useful under explicit scope notes, but they cannot be promoted as theorem-native.

- FRAME_ONLY and EXTENSION rows belong in support layers, appendices, narrower companion artifacts, or future work rather than in the positive body of every outward paper.
- REFUTED rows are excluded from the positive proof path. They may be mentioned only as negative audit history or as a falsifier record, not as live support.
- Frontier or hypothesis material may be discussed only under visibly labelled frontier language and may not be silently promoted by summary prose.

The public positive closure claim is scoped exactly to CORE_1_3_SCIENCE_ONLY. FRAME_ONLY rows may still close as frame-level PASS rows inside the synthesis surface, but they do not become standalone theorem-native empirical promotions. EXTENSION and REFUTED rows remain outside the public PASS scope for different reasons: EXTENSION rows are frontier/support lanes awaiting later promotion or rejection, while REFUTED rows are negative-audit lanes. Neither class can carry positive closure claims in this release. For avoidance of doubt, a FRAME_ONLY PASS row is excluded from the public positive closure claim unless a later release promotes it into CORE_1_3_SCIENCE_ONLY with the required theorem-native support.

This is not retreat. It is a condition of honest publication. The monograph can remain large without rhetorical inflation only if the reader can tell which rows are theorem-bearing, which are bounded, and which are support or frontier material.

2.4 Relation to journal-core and companion artifacts

Core 1.3 therefore adopts a one-way reader-facing publication logic:

Source-owned science corpus
plus canonical SPOT
$\Downarrow$
Master Monograph
$\Downarrow$
Derived outward artifacts

This diagram describes publication order, not the full provenance graph of the release package. Several editorial JSON surfaces, generated science chapters, and release matrices are direct SPOT projections or mirrors of other governing canonical surfaces. They do not become true merely because the master monograph cites them, and they are not descendants of the monograph alone. The monograph is the flagship reading object; the source-owned science corpus remains the authoring and witness locus, and the SPOT remains the canonical projection authority for release surfaces.

The journal-core article is narrower by design. Companion dossiers, matrices, and release surfaces are narrower or more operational by design. Those artifacts may compress, subset, or reorganise the flagship material for a specific task, but they may not enlarge the scientific claim, widen notation, or outrun the theorem-fate discipline fixed here.

The practical consequence is simple. If an outward reader-facing artifact says something stronger than the source corpus and SPOT authorise, the outward artifact is wrong, even if it is shorter, cleaner, or easier to market. The flagship boundary is the publication rule that enforces that ceiling; it is not an additional source of scientific authority.

2.5 Downstream projection rule

The same boundary stabilizes later use in review, collaboration, product, business, and public communication. Different channels may legitimately cite different parts of the flagship volume:

- the theorem route when bounded proof is the issue,

- the empirical route when promotion discipline is the issue,
- the practical route when use and comparison are the issue,
- the audit appendices when challenge and traceability are the issue.

What they may not do is invent a second scientific authority. Source witnessing, theorem fate, and publication boundary exist so that external use can remain concise without becoming interpretively lawless.

Bridge to the next chapter. With the publication boundary fixed, the next task is internal consistency: which statement classes the flagship volume uses, why the hierarchy runs through K_{12}, and how upper levels remain lawful without rhetorical inflation.

3 Core 1.3 Foundational Consistency Dossier

This chapter fixes the internal consistency contract of the flagship volume. Its purpose is not to replace the formal chapters with editorial commentary, but to make three things readable in one place:

- why the present hierarchy is fixed as K_0 through K_{12} rather than by an arbitrary cutoff;
- the full statement-class discipline for axiomatic, derived, empirical, structural, interpretive, and frontier claims;
- how the master monograph and the narrower journal-core extraction are related without silent drift in scope or notation.

3.1 Why the hierarchy extends through K_{12}

Core 1.2 already contains enough structure to motivate a release hierarchy larger than K_{10}. The point is not that higher labels are rhetorically attractive. The point is that the audited source corpus already distinguishes:

1. lower-level continua whose axes, thresholds, and flows are internal to a given model class;
2. theoretical and meta-theoretical continua that organise theories and model-spaces themselves;
3. a further stratum in which operators, meta-logics, and model-spaces become explicit objects of structural evolution;
4. and an upper bounding layer in which global coherence conditions across such higher-order structures must themselves be represented.

That is why Core 1.3 carries the public hierarchy through K_{12} rather than stopping at K_{10}. This is a release-level design rationale, not a standalone impossibility proof. The supporting route is the K-level doctrine in Section **??**, the theorem-navigation discussion in Section 4, and the worked preservation test in Section **??**. Those sections assign K_{11} to evolving meta-theoretical objects and K_{12} to global compatibility and integrability conditions. The present release does not claim a proof that no further extension is conceivable in principle; it claims only that K_{12} is the highest level lawfully carried by the current audited source corpus and therefore the last public level of this release.

3.2 Scientific role of the upper levels

Level K_{11}. K_{11} is the first level at which formal ontologies, operator systems, meta-logics, and model-spaces are themselves treated as structured continua subject to admissibility, transformation, and collapse conditions. This level is needed once one studies not only theories, but the evolution of the spaces in which theories and theory-building operators live.

Level K_{12}. K_{12} is not introduced as a mystical completion layer. It is the minimally specified upper coherence stratum required when one needs to discuss global compatibility, boundedness, and failure modes across families of K_{11}-objects. Its role is structural rather than triumphalist: it prevents the hierarchy from pretending that there is no formal problem above recursively evolving meta-theoretical spaces.

3.3 Statement classes

To keep the monograph readable without blurring scientific status, Core 1.3 uses the following proof-status vocabulary in this chapter. It is the local theorem and claim-scope taxonomy; Section 4 adds the downstream operational inspection burden for replay, comparator, falsifier, and escalation rules.

- **Axiomatic statements** fix primitive assumptions, admissibility conditions, or identity rules.

- **Derived statements** are theorem-bearing statements with explicit support dependence and proof obligations.
- **Empirical statements** require a standalone source-owned domain packet, canonical theorem-native trace, pinned data route, numerical parameter law, observable binding, locked held-out replay procedure, comparator audit, and explicit falsifier before promotion.
- **Structural statements** describe the organisation of the framework, the hierarchy, and the mapping between modules or levels.
- **Interpretive statements** explain what the formal structure means, which reader it is for, and what later projections may test.
- **Frontier statements** may remain in the research program, but may not enter positive outward science without formal promotion.

These local classes are not assumed to be disjoint. When a sentence could fit more than one class, its public status is assigned by semantic role and proof function rather than by surface phrasing:

1. an explicit contradiction of a promoted claim first flags a refutation-candidate or surface-contradiction case for SPOT audit;
2. empirical content is inspected by packet, data route, replay, comparator, and falsifier support before any prose label is trusted;
3. derived theorem content is inspected by its proof route and dependency graph;
4. admitted primitive assumptions remain axiomatic premises;
5. structural mapping and hierarchy claims remain structural even when they mention possible later use;
6. explanatory, audience-facing, or didactic sentences remain interpretive unless they assert theorem or empirical force;
7. unpromoted research-program proposals remain frontier statements;
8. pure structural or interpretive content keeps the SPOT-recorded `FRAME_ONLY` or `EXTENSION` class unless it is explicitly attached to evidence-bearing bridge material;
9. `BRIDGE_ONLY` is a stable evidence-support class for rows or packets that carry bridge support under explicit scope notes; it is not a theorem-native promotion queue.

Words such as “frontier”, “candidate”, or “projection” are therefore diagnostic signals, not the class assignment rule itself.

These prose classes are a local proof-status taxonomy, not a replacement for the canonical SPOT scientific-class and closure-verdict vocabulary. Release verdicts such as `PASS` and `FAIL_CLOSED` remain a separate verdict layer. The mapping is:

- Axiomatic statements are admitted premises or identity rules. They may sit inside a `THEOREM_NATIVE` route as named assumptions, but the axiom itself is not thereby converted into a proved theorem.
- Derived statements map to `THEOREM_NATIVE` when the support route is closed.
- Empirical statements map to `THEOREM_NATIVE` only after the source-owned packet, theorem-native trace, pinned data route, numerical parameter law, observable binding, locked held-out replay procedure, comparator audit, and falsifier gates pass.
- Structural and interpretive statements map to `FRAME_ONLY` or `EXTENSION` as recorded by SPOT unless the statement is attached to an evidence-bearing bridge row.

- Frontier statements map to `EXTENSION` until a later section or release promotes them through the same theorem-native or empirical gates used for other positive claims, or records an explicit refutation.
- Fail-closed bridge, frame, or extension lanes retain the class recorded by SPOT; closure failure by itself is not refutation.
- Statements map to `REFUTED` only when SPOT records an explicit refutation rather than a blocked promotion.

This chapter classifies the statement classes that affect theorem status, structural scope, interpretation, empirical promotion, and frontier boundaries. Operational route discipline is handled later in the theorem-roadmap and proof-machinery chapters. Editorial-bridge language is not a seventh proof class; it is an interpretive and release-consistency label for statements that connect the monograph, journal core, dossiers, and public package without adding theorem force. Operational statements are the later exact-match, held-out replay, statistical-threshold, tail-breach, comparator, falsifier, and escalation rules attached to empirical promotion. Structural statements are reviewed by scope, dependency, and source-boundary checks instead. Both kinds of control are handled after the statement class, numerical parameter law, and observable binding have already been fixed here, and neither becomes a seventh proof-status class.

The journal-core extraction is allowed to narrow this body, but not to blur these classes into one another. An interpretive sentence may explain a theorem; it may not replace the theorem. A structural remark may motivate a projection; it may not silently serve as proof of that projection.

3.4 Proof-bearing and nonclaim layers

The master monograph therefore separates three layers explicitly:

1. the inherited formal backbone that still bears the structure of the core theory;
2. the Core 1.3 repair layer that clarifies witness discipline, theorem fate, chronology, and publication boundary;
3. the bounded nonclaim layer that states what the present release does not yet defend, even when the broader research program still pursues it.

This separation matters for hostile review. A reviewer should be able to ask, for any sentence that sounds strong: is it axiomatic, derived, empirical, structural, interpretive, or frontier? If it is derived, what supports it? If it is empirical, what packet, data route, replay, comparator, and falsifier support it? If it is structural or interpretive, what exactly is it not claiming? If it is frontier, where is it visibly barred from positive release authority?

3.5 Relation between master monograph and journal core

The canonical SPOT is the scientific authority of the release, and the source-owned science corpus is the authoring and witness locus behind that projection. The master monograph is the flagship reader-facing reference because it carries the full theory, the reader ladder, the appendices, and the explicit publication boundary in one place. The *Journal Core* is the bounded article artifact. The *journal-core extraction* is the operation that selects one defensible theorem-bearing route for editorial inspection without pretending that the whole monograph is already compressed into that shorter artifact.

For that reason the extraction rule is one-way:

$$\text{Master Monograph} \longrightarrow \text{Journal Core},$$

with explicit preservation of notation, nonclaim boundaries, and theorem scope. The extraction procedure may omit detail. The Journal Core may not enlarge the claim. The public closure boundary remains the `CORE_1_3_SCIENCE_ONLY` scope stated in Section 2. `FRAME_ONLY` closures may be `PASS` as frame support, but they do not become standalone empirical promotions. BRIDGE_ONLY rows are scope-bounded evidence support rather than public promoted closure; they are not re-labelled as theorem-native PASS rows unless a separate source-owned theorem-native packet is created and passes the canonical gates. EXTENSION and REFUTED lanes remain outside public PASS authority.

3.6 Reader-facing consequence

This chapter exists because scientific rigor and readability do not pull in opposite directions here. The more explicit the dossier becomes about hierarchy, statement class, and proof boundary, the easier it becomes for three readers at once:

- the editor who needs a clean overview of what is actually defended;
- the formal reader who needs exact scope and dependency discipline;
- the non-specialist scientific reader who needs a coherent narrative entry point before diving into the full technical shell.

Bridge to the next chapter. With the statement classes, hierarchy boundary, and master-to-derivative relation fixed, the next chapter narrows further to one task only: the shortest lawful proof-navigation route through the load-bearing theorem spine.

4 Theorem Roadmap and Proof Navigation

This chapter is the clearest compact map of the proof-bearing route. Its task is to show where the load-bearing theorem spine lies, which statement classes carry which scientific burden, and where a hostile reviewer should probe first. It does not replace the formal chapters or the reader guide. The reader guide owns entry routes; this chapter owns proof navigation.

4.1 Load-bearing theorem spine

At the level of the bounded journal-core claim, the present release depends on a relatively small load-bearing spine even though the monograph itself remains much larger. The schematic route below is compact rather than proof-theoretic minimality in the strict sense: it keeps the residue and rebirth branch visible because the defended article-level theorem classifies post-collapse identity, not death alone. The spine can therefore be stated schematically as dependency notation rather than object-language implication. To avoid making the route look like a proof calculus, the support map is written as labelled review edges:

1. **Edge 1.** Axiom 3.2 together with Definitions 12.1, 12.3, and 12.4 supports Source Theorem 3, and Source Theorem 3 supports Lemma 1.
2. **Edge 2.** Axiom 3.3 together with Definition 12.6 supports Source Corollary 3.2, and Source Corollary 3.2 supports Lemma 3.
3. **Edge 3.** Lemma 1 together with Definition 12.5 supports Lemma 2.
4. **Edge 4.** Lemma 2 is the sole premise recorded for Theorem A in the canonical proof-obligation registry.

Definitions in this list are prerequisite nodes, not independent theorem premises. The word "supports" marks an admitted release support step rather than a primitive implication operator of OC.

Lemma 3 remains an admitted rebirth-control obligation in the surrounding roadmap, but it is not a premise of Theorem A in the canonical proof-obligation registry. Its load-bearing role is different: it supports the companion rebirth-control statement, Proposition B. Whenever a compressed article proof discusses the death–residue–rebirth package in one paragraph, Lemma 3 should be read as supporting Proposition B, while Theorem A itself remains the death-and-residue classification routed through Lemma 2.

Proposition B (article-local rebirth control). Within the article-level extraction, after K has lawfully collapsed so that $\Omega(K) = \emptyset$ and $k(K, t) = 0$, a later object K' is classified as rebirth if and only if K' is live, is numerically distinct from the dead original continuum K, appears either in the same-embedding branch or in an explicitly witnessed related-embedding branch, inherits at least one declared residue support element, and satisfies the relevant birth conditions. The identity rule supplies the reason this non-identity condition is forced rather than optional. This proposition is a companion control statement for the rebirth branch; it is not a premise of Theorem A.

In this roadmap, the *Source* prefix marks a witness projection from the audited public LaTeX source corpus and archival compiled PDF described in the source-audit appendix. Such rows are carried into Core 1.3 as source-witnessed support nodes and are inspected through the canonical proof-obligation registry; they are not newly asserted primitive theorems of the article-level extraction. The unprefixed lemmas, Theorem A, and Proposition B are the article-level derived rows assembled from that witnessed support. Theorem A uses the death-and-residue spine; Proposition B uses the rebirth-control branch.

Axioms, source theorems and corollaries, lemmas, and Theorem A carry proof-force nodes; definitions fix the terms used by those nodes.

The point of making that chain explicit is not to pretend that the whole framework is reducible to one diagram. The point is to show that the bounded claim has an auditable positive core rather than a cloud of motivational prose. Figure 1 also marks one excluded lift-heavy temptation route as a hostile-review boundary; it is reviewed precisely so that it cannot be mistaken for admitted support in the positive chain. The Corollary 3.2 / Lemma 3 branch is shown as an admitted rebirth-control branch for Proposition B beside the theorem route rather than as an antecedent of Theorem A. In this roadmap the related-embedding branch is represented by one auditable article-local witness, not by an exhaustive list of every admissible source-level related-embedding witness. This is the same branch denoted $\mathcal{M} \sim_{\text{res}} \mathcal{M}'$ in Section **??**: a named structure-preserving map $h : \mathcal{M} \to \mathcal{M}'$, a residue element $r \in R_{\text{res}}(K)$, admissible-region membership $h(r) \in \Omega(K') \subseteq \Omega_{\text{amb}}(\mathcal{M}')$, and preservation of the threshold and admissibility predicates on the residue-bearing substructure. Section **??** and the journal core state that witness test explicitly.

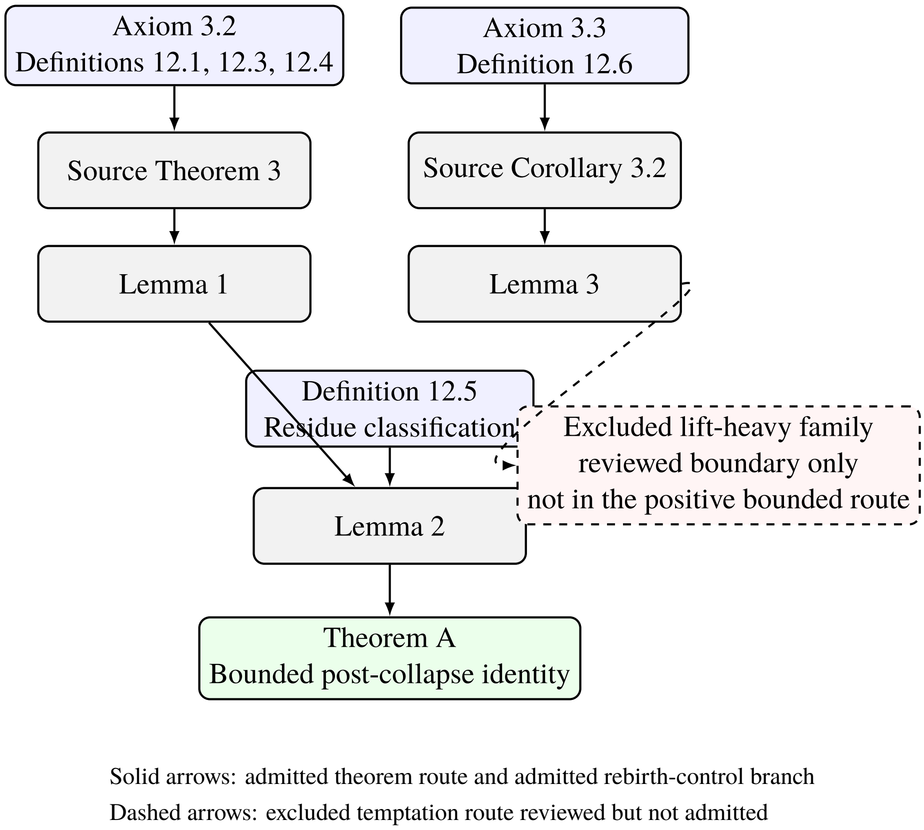

Figure 1: Load-bearing theorem spine of the bounded Core 1.3 claim. Solid arrows mark admitted support. The positive Theorem A route terminates at Lemma 2; the Lemma 3 branch remains visible as admitted rebirth-control support beside that route. The dashed branch marks an excluded lift-heavy route that is reviewed as a boundary condition rather than admitted into the positive proof. The larger monograph remains necessary because the surrounding chapters fix the ontology, operator shell, K-level doctrine, collapse grammar, empirical bar, and audit layers that keep this spine scientifically interpretable and externally citable.

4.2 What surrounds the spine

The theorem spine has a small proof dependency, but it is interpreted and audited inside the wider manuscript in five distinct ways:

1. the formal doctrine chapters fix admissibility, thresholds, operators, levels, branching, and falsifiability;
2. the consistency and source-boundary chapters fix statement class, theorem fate, and publication scope;

3. the proof-machinery chapter and appendix fix revision, comparator, compression, and proof-obligation discipline;
4. the operational and empirical chapters fix promotion, replay, and falsifier discipline;
5. the appendices keep the audit and atlas burden inside the same volume.

That is why the monograph is allowed to be larger than the defended article without becoming rhetorically loose. These surrounding chapters are publication and audit scaffolding, not additional premises of Theorem A.

4.3 Statement classes and burden

The table below is a proof-navigation map for reviewer use. It follows the statement-class taxonomy of Section 3. The proposition row is a navigation-only subdivision of derived statements, not a new proof-status class. Operational rules remain a downstream inspection burden rather than a separate proof-status class.

Statement class	Scientific burden	Where to inspect it
Axiomatic statements	Fix primitive admissibility, identity, or threshold conditions.	Axioms Appendix, Notation Appendix, and Sections **??**–**??**.
Derived theorems and lemmas	Carry the proof-bearing positive body.	Sections **??**, **??**, 4, Reviewer Navigation Matrix, and Proof Machinery Appendix.
Article-level propositions	Carry companion control statements that are derived from the same audited support but are not premises of Theorem A. Proposition B is the rebirth-control example.	Sections 4, **??**, and Journal Core Bridge Appendix.
Structural statements	Explain hierarchy, module layout, and relation between levels or artifacts.	Sections **??**–**??**, Section **??**, Section 2, and Section 3.
Interpretive statements	Explain meaning, nonclaims, and downstream boundaries.	Sections **??**, **??**, **??**, Source Audit Appendix, and Reviewer Navigation Matrix.
Empirical statements	Require a source-owned domain packet, theorem-native trace, numerical parameter law, observable binding, pinned data route, held-out replay, comparator audit, and falsifier before promotion.	Sections **??**–**??** and Appendices **??**–**??**.
Frontier statements	Mark lawful future work without positive release authority.	Section **??**, Appendix **??**, and the frontier rows in the source-owned science corpus.

Table 2: Statement classes and their inspection burden in Core 1.3. The table does not replace the formal chapters; it simply prevents proof, interpretation, empirical promotion, and frontier language from quietly changing roles. Replay, comparator, falsifier, and escalation rules are downstream control obligations attached to those classes.

All inspection targets in Table 2 are label-checked against the English master source. If a release wrapper changes one of those labels, the table must be regenerated or patched before the roadmap is treated as a mechanically followable audit route.

4.4 Shortest hostile-review checklist

If a reviewer wants the most efficient attack sequence, the logically hardest checks remain the following:

1. verify that collapse is tied to loss of admissible realization rather than to loose metaphor;

2. verify that the irreversibility rule forbids restoration of the original continuum as numerically the same entity;
3. verify that residue and rebirth are controlled classifications rather than narrative labels;
4. verify that no excluded lift-heavy theorem family is smuggled back into the positive bounded route;
5. verify that the journal-core extraction says nothing stronger than the source-owned science corpus and canonical SPOT permit; the flagship monograph is then checked only as a derived comparison surface;
6. verify that the K-level doctrine, especially K_{11} and K_{12}, is structurally justified rather than rhetorically inflated.

If the release fails there, later prose cannot rescue it. If it survives there, a slower full reading becomes worth the effort.

Bridge to the next chapter. With the proof route mapped, the next chapter changes mode deliberately. It no longer navigates the spine abstractly; it shows how the formal grammar should be read through worked examples, counterexamples, and bounded interpretive checks.

A Core 1.3 Source-Audit Appendix

This appendix records the principles by which the 1.3 release remains source-first while still operating as a publication program rather than as a static archive.

A.1 Why an appendix is still needed

The main body of the monograph is written to be read. The source-audit machinery is written to be checked. Those are related but non-identical tasks. If the full provenance burden were pushed into the main body, the book would become harder to read. If it were omitted entirely, the reader would need to trust a hidden workflow. The appendix therefore carries the discipline without replacing the main narrative.

A.2 Witness classes

The release uses several witness classes:

- the public LaTeX corpus as the source-native textual witness,
- the archival compiled PDF as the stable page-and-line witness,
- the bounded journal-core extraction as the outward review-focused derivative witness,
- and companion packets as support witnesses for chronology, theorem fate, and release packaging.

None of these witnesses should be mistaken for the others. The source witness is not the same as the release manifest. The release manifest is not the same as the theorem. The journal-core article is not the same as the monograph. The 1.3 publication line remains healthy only when these roles stay distinct.

A.3 Questions that the appendix answers

This appendix is meant to answer concrete review questions.

- From which public source stack is the present volume built?
- Which outward article is derived from it?
- Why is the monograph larger than the journal-core article?
- What justifies calling the volume 1.3 rather than simply republishing 1.2?
- How should later ESTRA and channel projections interpret the distinction between formal master, review article, and support packet?

The release is stronger when those questions can be answered inside the volume rather than by pointing the reader to an unrelated dashboard.

A.4 Current boundary

Core 1.3 should therefore be read as a source-first enlargement and clarification of the formal publication shell. The monograph does not pretend that every future projection is already proved. The journal core does not pretend to be the whole theory. The companion packets do not pretend to be the manuscript. That three-way distinction is the governing boundary of the current release.

B Reviewer Navigation Matrix and Audit Checklist

Start with the row that matches the question or concern, inspect the primary route, and use the verification column as the pass/fail test. This appendix uses three canonical artifact terms:

- *Master Monograph*: the English flagship volume. Later uses of *Monograph* are shorthand for this same artifact.
- *Journal Core*: the shorter bounded extraction from the monograph.
- *Release*: the outward package that also contains supporting surfaces, PDFs, assets, and companion dossiers.

Capitalisation marks the canonical glossary terms above; lowercase uses are generic prose shorthand unless explicitly styled as one of those terms. The following class labels are local definitions for this appendix rather than additional canonical glossary terms. Concrete artifact and witness classes are as follows:

- *Surfaces* are the release-bundled `editorial/*_latest.json` projections.
- *Companion dossiers* are the generated dossier packages under `editorial/dossier_packages/`.
- *Assets* are the released figures and tables under `assets/`.
- *Archival PDF witness*: the prior Core 1.2 PDF named in the journal-core provenance route in Appendix **??**: *Ontology of Continua — Core v1.2.0*.

B.1 Review questions, compact route, and decisive checks

Core theorem route and scope.

Reviewer question	**Primary route to inspect**	**Verification cue**
What is the central defended result?	Abstract; Sections **??**, 4	Theorem A: bounded post-collapse identity classification; no general lift theorem.
Where does the proof-bearing route begin?	Sections **??**, 4	Verify that Axiom 3.2 with Definitions 12.1, 12.3, and 12.4 supports Source Theorem 3; that Source Theorem 3 supports Lemma 1; that Lemma 1 with Definition 12.5 supports Lemma 2; and that Lemma 2 supports Theorem A. Inspect Axiom 3.3, Definition 12.6, and Source Corollary 3.2 separately as the companion rebirth-control branch.
Is collapse defined formally or rhetorically?	Sections **??**, **??**	Admissible-state emptiness, $k \to 0$, and threshold failure.
What exactly is residue?	Sections **??**, **??**	Post-collapse structural trace; no continuing live identity.
What exactly is rebirth?	Sections **??**, **??**	Later live continuum grounded in residue; numerically new.
Why is irreversibility necessary?	Sections **??**, 4	Blocks same-entity restoration after lawful death.
Could the main theorem depend on an excluded lift-heavy theorem family?	Section 4; Appendix **??**	No excluded lift-heavy theorem family is used in the bounded proof route.
Why is the monograph larger than the journal core?	Section 2; Appendix **??**	Didactic shell, formal architecture, and repair logic live here.
Why is the journal core smaller without becoming weaker?	Appendix **??**	One-way extraction; theorem scope unchanged.
Why is the hierarchy fixed as $K_0, \dots, K_{12}$?	Sections 3, **??**, **??**	Upper levels carry substantive higher-order roles.

Audit, provenance, and review posture.

Reviewer question	Primary route to inspect	Verification cue
What would falsify the upper-level doctrine?	Sections ??, ??	Lossless reduction of K_{11} and K_{12} to K_{10}.
How do proof-oriented statement classes differ?	Sections 3, 4	Axiomatic, derived, structural, interpretive, empirical, and frontier rows keep separate burdens; editorial-bridge language remains packaging support rather than a proof class.
What is merely interpretive?	Sections ??, 2, ??	Pass only if interpretive language stays outside the proof route and is not promoted into theorem, replay, or practical-use authority.
What is the exact source base?	Section 2; Appendix A	Source corpus, SPOT surface, and archival PDF witness are separate objects.
Why is ORCID foregrounded?	Title page; Section 2	Stable scholarly identity.
What is the role of companion dossiers?	Section 2; Appendix ??	Support and provenance exposure, not replacement authority.
How should a non-specialist enter the monograph?	Sections 1, ??	Didactic ladder before proof audit.
Where are the most likely points of critique?	Section 4; Appendix ??	Collapse semantics, irreversibility, upper levels, extraction discipline.
What would count as a decisive release failure?	Sections ??, ??, 2	Continued identity after admissible-state loss, or reliance on an excluded theorem family.
What remains outside scope?	Sections ??, 2; Appendix ??	Pass only if excluded theorem families, frontier rows, and projection targets remain outside the positive proof body.

B.2 Compact reading order by reviewer role

Rapid critical reviewer. Read the abstract, theorem roadmap, relevant theorems and corollaries, the worked examples, and the journal-core bridge appendix. Return to this matrix only to choose a new route after that first pass. This route should reveal immediately whether the release is too large, too narrow, or too broad in scope.

Formal mathematical reviewer. Read the model, results, axioms, operators, collapse/rebirth material, hierarchy chapter, notation appendix, and source-audit chapter. The decisive question is whether the derived bounded claim can be reconstructed from the stated source-native rows without informal supplementation.

Generalist scientific reviewer. Read the introduction, reader guide, theorem roadmap, worked examples, discussion, conclusion, and then return to the formal chapters selectively. The decisive question here is whether the volume explains itself honestly enough that scientific judgment can precede local notation mastery.

C Unified Synthesis Support Dossiers

The main unified-synthesis section remains the canonical synthesis unit in the numbered Core 1.3 sequence of the manuscript. The material gathered here is retained as level-wise evidence support: it keeps compact K-anchored theorem-support, empirical, falsifier, and packet dossiers available inside the flagship volume while omitting the synthesis-section introduction and release-gate prose. Section **??** uses the generated synthesis surface for the main K-level exposition. This appendix is the support-dossier wrapper: it gives readers a single inspection location for the level-wise support files without turning them into a second synthesis chapter. The main chapter states the synthesis, while this appendix carries the dossier-oriented inspection copy. Compatibility support files are not input directly by the English platinum master. A build variant must therefore choose either the compatibility synthesis-support master or this appendix wrapper, never both. The full DFS compatibility stream still records those compatibility entries for auditability, but the platinum master does not input the compatibility synthesis-support master. The constants-and-parameters apparatus is part of the Appendix **??** data apparatus and is addressable there as Appendix **??**; this support appendix does not create a second synthesis or a second top-level constants appendix. Source stamp: `releases/oc_core_1_3/editorial/OC_CORE_1_3_UNIFIED_SYNTHESIS_latest.json`; current repository SHA and UTC generation timestamp are recorded inside that JSON surface so this appendix does not freeze stale build metadata.

C.1 Level K0: structural conditions for any universe

Current closure status: `PASS`. Promotion blockers: none.

Theorem claim. K0 fixes the non-empty admissible meta-domain required for any lawful continuum and therefore bounds every later K-level before empirical specialization.

Operator and K-level binding. The local K0 root process glyphs Ψ_0, Φ_0, and Λ_0 fix distinction generation, relational reconfiguration, and compositional assembly; compact synthesis aliases F_0, Q_0, and U_0 label the corresponding root constraints only where they are defined locally. Kernel refs: `content/03_model.tex`, `content/10_klevels_full.tex`, `content/11_operators_full.tex`, `APPENDIX_Q_SYNTHESIS_DOSSIER_K0`.

Parameter law.

$$\mu(\Omega(K_0)) > 0, \quad \forall x \in \{1, \ldots, 12\} : \text{DoF}(M_x) > 0 \text{ and } \frac{\text{DoF}(K_x)}{\text{DoF}(M_x)} \leq 1, \quad C_{\text{nt}} \geq 1$$

Observable binding and units. Meta-admissibility, meta-space compatibility, and the presence of at least one non-trivial lawful cycle. Observable ids: `K0::ADMISSIBLE_STATE_MEASURE`, `K0::META_SPACE_COMPATIBILITY_RATIO`, `K0::NONTRIVIAL_CYCLE_PRESENCE`, `MATHEMATICS::EPSILON_STAR`, `MATHEMATICS::MINIMAL_DENOMINATOR`, `MATHEMATICS::MINIMAL_AUGMENTATION_RANK`, `MATHEMATICS::COMPONENT_FIBER_COUNT`, `MATHEMATICS::MOVE_CONNECTIVITY_OR_DEGREE_BOUND`, `MATHEMATICS::STRICT_TERMINALIZED_TOTAL`, `MATHEMATICS::COUNTEREXAMPLE_EXECUTED_TOTAL`, `MATHEMATICS::PYTHON_EVIDENCE_EXECUTED_TOTAL`.

Numerical instantiation and prediction table. The table below is a promoted subset of the numerical backing rows in Appendix **??**.

Quantity	Symbol	Value / range	Units or note
Admissible-state measure floor	$\mu(\Omega(K_0))$	> 0	dimensionless lower bound

Quantity	Symbol	Value / range	Units or note
Meta-space compatibility ratio for each promoted level	$\mathrm{DoF}(K_x)/\mathrm{DoF}(M_x)$	≤ 1	dimensionless ratio; applies for each x in 1,...,12
Non-trivial cycle presence	C_{nt}	1	present

Falsifier and collapse boundary. Collapse occurs if the admissible set degenerates, if a later K-level exceeds its meta-space, or if the structural substrate loses its last non-trivial lawful cycle. Falsifier refs: `APPENDIX_Q_SYNTHESIS_DOSSIER_K0`, `content/falsifiability/falsifiability_master.tex`, `content/falsifiability/falsifiability_k0.tex`, `content/falsifiability/falsifiability_k9.tex`, `content/falsifiability/falsifiability_k10.tex`, `content/falsifiability/falsifiability_k11.tex`, `content/falsifiability/falsifiability_k12.tex`.

Closed-domain bindings. `MATHEMATICS`: class `THEOREM_NATIVE`, verdict `PASS`, held-out cases `0`, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/mathematics`. Metrics: `strict_campaign_acceptance_status`=PASS_31_OF_31_TERMINALIZED, `strict_terminalized_total`=31.

C.2 Level K1: minimal observable distinctions and energetic axes

Current closure status: `PASS`. Promotion blockers: none.

Theorem claim. K1 introduces resolvable energetic and temporal distinctions while preserving the admissible-state floor inherited from K0.

Operator and K-level binding. F, G, and U bind contradiction load to resolvable energetic distinctions, monotonic update order, and a non-degenerate observable region. Kernel refs: `content/03_model.tex`, `content/10_klevels_full.tex`, `content/11_operators_full.tex`, `APPENDIX_Q_SYNTHESIS_DOSSIER_K1`.

Parameter law.

$$\Delta E \geq \Theta_{\Delta E}, \quad \frac{dt}{d\tau} > 0, \quad \mu(\Omega(K_1)) > 0$$

Observable binding and units. The first observable regime is tracked by energy-gap resolution, monotonic update ordering, and positive configuration-space measure. Observable ids: `K1::ENERGY_GAP_THRESHOLD`, `K1::TEMPORAL_MONOTONICITY`, `K1::CONFIGURATION_MEASURE`, `PHYSICS::FINE_STRUCTURE_CONSTANT_RESIDUAL`, `PHYSICS::HYDROGEN_BALMER_LINE_RESIDUAL`, `PHYSICS::TRANSPORT_SCALING_RESIDUAL`.

Numerical instantiation and prediction table. The table below is a promoted subset of the numerical backing rows in Appendix **??**.

Quantity	Symbol	Value / range	Units or note
Minimal resolvable energy scale	$\Theta_{\Delta E}$	10^{-34}–10^{-33}	J
Temporal monotonicity	$dt/d\tau$	> 0	ordered update constraint

Quantity	Symbol	Value / range	Units or note
Configuration-space measure floor	$\mu(\Omega(K_1))$	> 0	dimensionless lower bound

Falsifier and collapse boundary. Collapse begins when energetic distinctions are no longer resolvable, temporal order breaks, or the first observable state space degenerates. Falsifier refs: `APPENDIX_Q_SYNTHESIS_DOSSIER_K1`, `content/falsifiability/falsifiability_master.tex`, `content/falsifiability/falsifiability_k1.tex`, `content/falsifiability/falsifiability_k2.tex`, `content/falsifiability/falsifiability_k9.tex`, `content/falsifiability/falsifiability_k10.tex`.

Closed-domain bindings. `PHYSICS`: class `THEOREM_NATIVE`, verdict `PASS`, held-out cases `20`, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/physics`. Metrics: `covered_case_total`=30, `covered_held_out_case_total`=20, `normalized_error_mean_abs`=0.188136, `normalized_error_p95`=0.64778, `normalized_error_max`=1.48, `tail_breach_count`=0.

C.3 Level K2: fields, phases, and large-scale structure

Current closure status: `PASS`. Promotion blockers: none.

Theorem claim. K2 binds field configurations, expansion history, and phase thresholds into the first large-scale physical continuum admissible under the kernel operators.

Operator and K-level binding. F, H, and R constrain field admissibility, phase transitions, and large-scale stabilization margins inside the physical continuum. Kernel refs: `content/03_model.tex`, `content/10_klevels_full.tex`, `content/11_operators_full.tex`, `APPENDIX_Q_SYNTHESIS_DOSSIER_K2`.

Parameter law.

$$H_0 > 0, \quad 0 < \Omega_\mathrm{m} < 1, \quad |\Omega_k| < 2 \times 10^{-3}, \quad \mathcal{T}_\mathrm{phase}(t) \in \{T_c^\mathrm{EW}, T_c^\mathrm{QCD}\}$$

Observable binding and units. Expansion rate, curvature, cosmological density fractions, and critical phase windows tied to physical observables and replay packets. Observable ids: `K2::EXPANSION_RATE`, `K2::CURVATURE_BOUND`, `K2::PHASE_TRANSITION_WINDOW`, `CHEMISTRY::GAS_PHASE_ENTHALPY_RESIDUAL`, `CHEMISTRY::SPECTRAL_TRANSITION_RESIDUAL`, `CHEMISTRY::KINETIC_OR_EQUILIBRIUM_RESIDUAL`, `PHYSICS::FINE_STRUCTURE_CONSTANT_RESIDUAL`, `PHYSICS::HYDROGEN_BALMER_LINE_RESIDUAL`, `PHYSICS::TRANSPORT_SCALING_RESIDUAL`.

Numerical instantiation and prediction table. The table below is a promoted subset of the numerical backing rows in Appendix **??**.

Quantity	Symbol	Value / range	Units or note
Hubble parameter today	H_0	67.4 ± 0.5	$\mathrm{km\,s^{-1}\,Mpc^{-1}}$
Matter density fraction	Ω_m	0.315 ± 0.007	dimensionless
Critical phase windows	T_c^EW, T_c^QCD	$\sim$ 100 GeV; 150–170 MeV	closed phase set

Quantity	Symbol	Value / range	Units or note
Curvature bound	Ω_k	$\lvert\Omega_k\rvert < 2 \times 10^{-3}$	dimensionless

Falsifier and collapse boundary. K2 collapses when curvature exits the admissible band, phase windows fail to close lawfully, or large-scale physical observables lose replayable stabilization. Falsifier refs: `APPENDIX_Q_SYNTHESIS_DOSSIER_K2`, `content/falsifiability/falsifiability_master.tex`, `content/falsifiability/falsifiability_k2.tex`, `content/falsifiability/falsifiability_k3.tex`, `content/falsifiability/falsifiability_k4.tex`, `content/falsifiability/falsifiability_k6.tex`, `content/falsifiability/falsifiability_k1.tex`, `content/falsifiability/falsifiability_k9.tex`, `content/falsifiability/falsifiability_k10.tex`.

Closed-domain bindings. `CHEMISTRY`: class `THEOREM_NATIVE`, verdict `PASS`, held-out cases `19`, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/chemistry`. Metrics: `covered_case_total`=30, `covered_held_out_case_total`=19, `normalized_error_mean_abs`=0.533285, `normalized_error_p95`=1.0, `normalized_error_max`=1.597772, `tail_breach_count`=0. `DRT`: class `REFUTED`, verdict `FAIL_CLOSED`, held-out cases `0`, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `none`. Metrics: none. `PHYSICS`: class `THEOREM_NATIVE`, verdict `PASS`, held-out cases `20`, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/physics`. Metrics: `covered_case_total`=30, `covered_held_out_case_total`=20, `normalized_error_mean_abs`=0.188136, `normalized_error_p95`=0.64778, `normalized_error_max`=1.48, `tail_breach_count`=0.

C.4 Level K3: molecular organization and chemical bonding

Current closure status: `PASS`. Promotion blockers: none.

Theorem claim. K3 packages bounded molecular organization through lawful bond-length, bond-energy, and activation-energy envelopes derived from lower-level admissibility.

Operator and K-level binding. G, H, and R bind composition-sensitive bonding, activation barriers, and molecular-scale stabilization windows. Kernel refs: `content/03_model.tex`, `content/10_klevels_full.tex`, `content/11_operators_full.tex`, `APPENDIX_Q_SYNTHESIS_DOSSIER_K3`.

Parameter law.

$$r_{\text{cov}} \in [0.1, 0.2]\,\text{nm}, \quad E_{\text{bond}} \in [1, 5]\,\text{eV}, \quad E^{\ddagger} \in [0.1, 1]\,\text{eV}$$

Observable binding and units. Bond lengths, dissociation energies, dielectric context, and activation barriers for bounded chemistry lanes. Observable ids: `K3::BOND_LENGTH`, `K3::BOND_ENERGY`, `K3::ACTIVATION_ENERGY`, `CHEMISTRY::GAS_PHASE_ENTHALPY_RESIDUAL`, `CHEMISTRY::SPECTRAL_TRANSITION_RESIDUAL`, `CHEMISTRY::KINETIC_OR_EQUILIBRIUM_RESIDUAL`.

Numerical instantiation and prediction table. The table below is a promoted subset of the numerical backing rows in Appendix **??**.

Quantity	Symbol	Value / range	Units or note
Bond length (covalent)	r_{cov}	0.1–0.2	nm
Bond dissociation energy	E_{bond}	1–5	eV

Quantity	Symbol	Value / range	Units or note
Reaction activation energy	$E^{\ddagger}$	0.1–1	eV

Falsifier and collapse boundary. K3 collapses when bond- and barrier-level regularities cannot be bounded by a non-degenerate parameter law or when chemical observables drift outside admissible molecular windows. Falsifier refs: `APPENDIX_Q_SYNTHESIS_DOSSIER_K3`, `content/falsifiability/falsifiability_master.tex`, `content/falsifiability/falsifiability_k2.tex`, `content/falsifiability/falsifiability_k3.tex`, `content/falsifiability/falsifiability_k4.tex`.

Closed-domain bindings. `CHEMISTRY`: class `THEOREM_NATIVE`, verdict `PASS`, held-out cases `19`, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/chemistry`. Metrics: `covered_case_total`=30, `covered_held_out_case_total`=19, `normalized_error_mean_abs`=0.533285, `normalized_error_p95`=1.0, `normalized_error_max`=1.597772, `tail_breach_count`=0.

C.5 Level K4: compartmental and membrane thresholds

Current closure status: `PASS`. Promotion blockers: none.

Theorem claim. K4 closes the first compartmental organization packet by binding membrane thickness, bending stability, and gradient maintenance to lawful biological thresholds.

Operator and K-level binding. H, R, and S couple membrane stability, osmotic admissibility, and retained gradients into a bounded protocellular regime. Kernel refs: `content/03_model.tex`, `content/10_klevels_full.tex`, `content/11_operators_full.tex`, `APPENDIX_Q_SYNTHESIS_DOSSIER_K4`.

Parameter law.

$$d_{\mathrm{mem}} \in [3,5]\,\mathrm{nm}, \quad \kappa \in [10,25]\,k_{\mathrm{B}}T, \quad \Delta pH \in [0.5,2]$$
$$\gamma_{\mathrm{edge}} \in [5,20]\,\mathrm{pN}, \quad \Pi \in [10^5,10^6]\,\mathrm{Pa}, \quad P_i \in [10^{-12},10^{-8}]\,\mathrm{m/s}$$

Observable binding and units. Membrane thickness, bending modulus, osmotic pressure, proton gradient, and permeability lanes for compartmental organization. Observable ids: `K4::MEMBRANE_THICKNESS`, `K4::BENDING_MODULUS`, `K4::PROTON_GRADIENT`, `K4::LINE_TENSION`, `K4::OSMOTIC_PRESSURE_SCALE`, `K4::PERMEABILITY_RANGE`, `BIOLOGY::EXPRESSION_SIGNATURE_RESIDUAL`, `BIOLOGY::STATE_TRANSITION_ORDER_ERROR`, `BIOLOGY::RESPONSE_ONSET_RESIDUAL`, `CHEMISTRY::GAS_PHASE_ENTHALPY_RESIDUAL`, `CHEMISTRY::SPECTRAL_TRANSITION_RESIDUAL`, `CHEMISTRY::KINETIC_OR_EQUILIBRIUM_RESIDUAL`.

Numerical instantiation and prediction table. The table below is a promoted subset of the numerical backing rows in Appendix **??**.

Quantity	Symbol	Value / range	Units or note
Membrane thickness	d_{mem}	3–5	nm
Membrane bending modulus	κ	10–25	$k_{\mathrm{B}}T$
Typical proton gradient	ΔpH	0.5–2	pH units

Quantity	Symbol	Value / range	Units or note
Line tension (edge)	γ_{edge}	5–20	pN
Osmotic pressure scale	Π	10^5–10^6	Pa
Permeability range	P_i	10^{-12}–10^{-8}	m/s

Falsifier and collapse boundary. K4 collapses if compartments cannot retain a lawful gradient, if membranes lose structural stability, or if permeability escapes the admissible packet. Falsifier refs: `APPENDIX_Q_SYNTHESIS_DOSSIER_K4`, `content/falsifiability/falsifiability_master.tex`, `content/falsifiability/falsifiability_k4.tex`, `content/falsifiability/falsifiability_k5.tex`, `content/falsifiability/falsifiability_k6.tex`, `content/falsifiability/falsifiability_k7.tex`, `content/falsifiability/falsifiability_k2.tex`, `content/falsifiability/falsifiability_k3.tex`.

Closed-domain bindings. `BIOLOGY`: class `THEOREM_NATIVE`, verdict `PASS`, held-out cases `16`, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/biology`. Metrics: `covered_case_total`=30, `covered_held_out_case_total`=16, `normalized_error_mean_abs`=0.444222, `normalized_error_p95`=1.030033, `normalized_error_max`=1.438582, `tail_breach_count`=0. `CHEMISTRY`: class `THEOREM_NATIVE`, verdict `PASS`, held-out cases `19`, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/chemistry`. Metrics: `covered_case_total`=30, `covered_held_out_case_total`=19, `normalized_error_mean_abs`=0.533285, `normalized_error_p95`=1.0, `normalized_error_max`=1.597772, `tail_breach_count`=0.

C.6 Level K5: excitable and early neural continua

Current closure status: `PASS`. Promotion blockers: none.

Theorem claim. K5 binds excitability, refractory timing, and membrane potential maintenance into the first lawful neural packet.

Operator and K-level binding. F, G, and S stabilize excitable-state transitions, ion-channel gating, and refractory boundaries in bounded neural lanes. Kernel refs: `content/03_model.tex`, `content/10_klevels_full.tex`, `content/11_operators_full.tex`, `APPENDIX_Q_SYNTHESIS_DOSSIER_K5`.

Parameter law.

$$C_m = 1\,\mu\text{F/cm}^2, \quad V_{\text{rest}} \in [-75, -60]\,\text{mV}, \quad \tau_{\text{ref}} \in [2, 5]\,\text{ms}$$

Observable binding and units. Membrane capacitance, resting potential, channel conductance, and refractory timing for bounded excitable systems. Observable ids: `K5::MEMBRANE_CAPACITANCE`, `K5::RESTING_POTENTIAL`, `K5::REFRACTORY_TIME`, `BIOLOGY::EXPRESSION_SIGNATURE_RESIDUAL`, `BIOLOGY::STATE_TRANSITION_ORDER_ERROR`, `BIOLOGY::RESPONSE_ONSET_RESIDUAL`.

Numerical instantiation and prediction table. The table below is a promoted subset of the numerical backing rows in Appendix **??**.

Quantity	Symbol	Value / range	Units or note
Membrane capacitance	C_m	1	μF/cm^2
Resting membrane potential	V_{rest}	$-75--60$	mV
Refractory time	τ_{ref}	2–5	ms

Falsifier and collapse boundary. K5 collapses when excitability loses bounded thresholds, refractory timing fails, or the neural packet cannot survive counterexample pressure across datasets. Falsifier refs: `APPENDIX_Q_SYNTHESIS_DOSSIER_K5`, `content/falsifiability/falsifiability_master.tex`, `content/falsifiability/falsifiability_k4.tex`, `content/falsifiability/falsifiability_k5.tex`, `content/falsifiability/falsifiability_k6.tex`, `content/falsifiability/falsifiability_k7.tex`.

Closed-domain bindings. `BIOLOGY`: class `THEOREM_NATIVE`, verdict `PASS`, held-out cases `16`, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/biology`. Metrics: `covered_case_total`=30, `covered_held_out_case_total`=16, `normalized_error_mean_abs`=0.444222, `normalized_error_p95`=1.030033, `normalized_error_max`=1.438582, `tail_breach_count`=0.

C.7 Level K6: cognitive prediction and representation

Current closure status: `PASS`. Promotion blockers: none.

Theorem claim. K6 formalizes bounded cognitive prediction and representation by linking prediction-error thresholds, working-memory span, and adaptation strength.

Operator and K-level binding. Q, R, and S bind representation stability, prediction thresholds, and adaptive update constraints in bounded cognitive lanes. Kernel refs: `content/03_model.tex`, `content/10_klevels_full.tex`, `content/11_operators_full.tex`, `APPENDIX_Q_SYNTHESIS_DOSSIER_K6`.

Parameter law.

$$\Theta_{\text{pred}} \in [0.05, 0.15], \quad M_{\text{WM}} \in [3, 7], \quad \eta \in [10^{-3}, 10^{-1}]$$

Observable binding and units. Prediction error, working-memory span, and adaptation strength for bounded cognitive organization. Observable ids: `K6::PREDICTION_ERROR_THRESHOLD`, `K6::WORKING_MEMORY_SPAN`, `K6::HEBBIAN_STRENGTH`, `BIOLOGY::EXPRESSION_SIGNATURE_RESIDUAL`, `BIOLOGY::STATE_TRANSITION_ORDER_ERROR`, `BIOLOGY::RESPONSE_ONSET_RESIDUAL`.

Numerical instantiation and prediction table. The table below is a promoted subset of the numerical backing rows in Appendix **??**.

Quantity	Symbol	Value / range	Units or note
Prediction-error threshold	Θ_{pred}	0.05–0.15	normalized units
Working-memory span	M_{WM}	3–7	bound items
Hebbian strength coefficient	η	10^{-3}–10^{-1}	dimensionless

Falsifier and collapse boundary. K6 collapses when predictive stabilization fails across bounded cognitive observables or when representation thresholds require hidden freedom. Falsifier refs: `APPENDIX_Q_SYNTHESIS_DOSSIER_K6`, `content/falsifiability/falsifiability_master.tex`, `content/falsifiability/falsifiability_k4.tex`, `content/falsifiability/falsifiability_k5.tex`, `content/falsifiability/falsifiability_k6.tex`, `content/falsifiability/falsifiability_k7.tex`, `content/falsifiability/falsifiability_k2.tex`.

Closed-domain bindings. `BIOLOGY`: class `THEOREM_NATIVE`, verdict `PASS`, held-out cases `16`, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/biology`. Metrics: `covered_case_total`=30, `covered_held_out_case_total`=16, `normalized_error_mean_abs`=0.444222, `normalized_error_p95`=1.030033, `normalized_error_max`=1.438582, `tail_breach_count`=0. `DRT`: class `REFUTED`, verdict `FAIL_CLOSED`, held-out cases `0`, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `none`. Metrics: none.

C.8 Level K7: social coordination and group stability

Current closure status: `PASS`. Promotion blockers: none.

Theorem claim. K7 closes bounded social coordination through group size, communication bandwidth, and coordination thresholds.

Operator and K-level binding. R, S, and U bind coordination thresholds, communication throughput, and conflict costs into admissible social trajectories. Kernel refs: `content/03_model.tex`, `content/10_klevels_full.tex`, `content/11_operators_full.tex`, `APPENDIX_Q_SYNTHESIS_DOSSIER_K7`.

Parameter law.

$$N_s \in [120, 180], \quad J_{\text{comm}} \in [1, 5]\,\text{bits/s}, \quad \Theta_{\text{coord}} \in [0.2, 0.4]$$
$$C_{\text{conf}} \in [0.1, 1]$$

Observable binding and units. Group stability size, communication throughput, coordination threshold, and conflict cost inside social coordination continua. Observable ids: `K7::GROUP_STABILITY_SIZE`, `K7::COMMUNICATION_BANDWIDTH`, `K7::COORDINATION_THRESHOLD`, `K7::CONFLICT_COST_SCALE`, `BIOLOGY::EXPRESSION_SIGNATURE_RESIDUAL`, `BIOLOGY::STATE_TRANSITION_ORDER_ERROR`, `BIOLOGY::RESPONSE_ONSET_RESIDUAL`.

Numerical instantiation and prediction table. The table below is a promoted subset of the numerical backing rows in Appendix **??**.

Quantity	Symbol	Value / range	Units or note
Group stability size	N_s	120–180	agents
Information-flow bandwidth per agent	J_{comm}	1–5	bits/s
Coordination threshold	Θ_{coord}	0.2–0.4	dimensionless
Conflict cost scale	C_{conf}	0.1–1	arbitrary units

Falsifier and collapse boundary. K7 collapses when communication throughput and coordination thresholds cannot stabilize social group trajectories without hidden rescue assumptions. Falsifier refs: APPENDIX_Q_SYNTHESIS_DOSSIER_K7, content/falsifiability/falsifiability_master.tex, content/falsifiability/falsifiability_k4.tex, content/falsifiability/falsifiability_k5.tex, content/falsifiability/falsifiability_k6.tex, content/falsifiability/falsifiability_k7.tex.

Closed-domain bindings. BIOLOGY: class THEOREM_NATIVE, verdict PASS, held-out cases 16, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/biology. Metrics: covered_case_total=30, covered_held_out_case_total=16, normalized_error_mean_abs=0.444222, normalized_error_p95=1.030033, normalized_error_max=1.438582, tail_breach_count=0.

C.9 Level K8: civilizational flows and infrastructural stability

Current closure status: PASS. Promotion blockers: none.

Theorem claim. K8 binds civilizational energy density, renewal timing, and repair-decay balance into bounded macro-process trajectories and regime-shift packets.

Operator and K-level binding. H, R, and U bind throughput, repair-decay balance, and regime-shift boundaries in bounded systems-level observables. Kernel refs: content/03_model.tex, content/10_klevels_full.tex, content/11_operators_full.tex, APPENDIX_Q_SYNTHESIS_DOSSIER_K8.

Parameter law.

$$E_{\text{dens}} \in [10^2, 10^3]\,\mathrm{W/m^2}, \quad \tau_{\text{infra}} \in [20, 50]\ \text{years}, \quad R_{\text{rep}} \in [0.8, 1.2]$$
$$\Theta_I \in [0.1, 0.3]$$

Observable binding and units. Energy density, infrastructure renewal, information coherence, and repair-decay balance in bounded civilizational lanes. Observable ids: K8::ENERGY_CONSUMPTION_DENSITY, K8::INFRASTRUCTURE_RENEWAL_TIME, K8::REPAIR_TO_DECAY_RATIO, K8::INFORMATION_COHERENCE_THRESHOLD, SYSTEMS_CIVILIZATIONAL_PROJECTION::POPULATION_TRAJECTORY_RESIDUAL, SYSTEMS_CIVILIZATIONAL_PROJECTION::GDP_GROWTH_TURNING_POINT_ERROR, SYSTEMS_CIVILIZATIONAL_PROJECTION::PROCESS_SYSTEM_REGIME_SCORE.

Numerical instantiation and prediction table. The table below is a promoted subset of the numerical backing rows in Appendix **??**.

Quantity	Symbol	Value / range	Units or note
Energy consumption density	E_{dens}	10^2–10^3	$\mathrm{W/m^2}$
Infrastructure renewal time	τ_{infra}	20–50	years
Information-coherence threshold	Θ_I	0.1–0.3	dimensionless
Repair-to-decay ratio	R_{rep}	0.8–1.2	dimensionless

Falsifier and collapse boundary. K8 collapses when macro-trajectories lose bounded turning-point fidelity, when repair-decay balance drifts outside its band, or when regime-shift packets require discretionary knobs. Falsifier refs: `APPENDIX_Q_SYNTHESIS_DOSSIER_K8`, `content/falsifiability/falsifiability_master.tex`, `content/falsifiability/falsifiability_k8.tex`.

Closed-domain bindings. `SYSTEMS_CIVILIZATIONAL_PROJECTION`: class `THEOREM_NATIVE`, verdict `PASS`, held-out cases `18`, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/systems_civilizational_projection`. Metrics: `covered_case_total`=30, `covered_held_out_case_total`=18, `normalized_error_mean_abs`=0.56097, `normalized_error_p95`=1.924704, `normalized_error_max`=2.25395, `tail_breach_count`=0.

C.10 Level K9: knowledge systems and scientific memory

Current closure status: `PASS`. Promotion blockers: none.

Theorem claim. K9 formalizes bounded scientific-knowledge continua through anomaly accumulation, paradigm coherence, and model-validation bandwidth.

Operator and K-level binding. Q, R, and U bind anomaly handling, paradigm coherence, and memory retention inside knowledge-system packets. Kernel refs: `content/03_model.tex`, `content/10_klevels_full.tex`, `content/11_operators_full.tex`, `APPENDIX_Q_SYNTHESIS_DOSSIER_K9`.

Parameter law.

$$\Theta_{\text{par}} \in [0.15, 0.25], \quad J_{\text{anom}} \in [10^{-4}, 10^{-2}], \quad \tau_{\text{mem}} \in [30, 200] \text{ years}$$
$$B_{\text{val}} \in [0.1, 1]$$

Observable binding and units. Paradigm coherence, anomaly accumulation, validation bandwidth, and scientific-memory half-life. Observable ids: `K9::PARADIGM_COHERENCE_THRESHOLD`, `K9::ANOMALY_ACCUMULATION_RATE`, `K9::SCIENTIFIC_MEMORY_HALF_LIFE`, `K9::MODEL_VALIDATION_BANDWIDTH`, `MATHEMATICS::EPSILON_STAR`, `MATHEMATICS::MINIMAL_DENOMINATOR`, `MATHEMATICS::MINIMAL_AUGMENTATION_RANK`, `MATHEMATICS::COMPONENT_FIBER_COUNT`, `MATHEMATICS::MOVE_CONNECTIVITY_OR_DEGREE_BOUND`, `MATHEMATICS::STRICT_TERMINALIZED_TOTAL`, `MATHEMATICS::COUNTEREXAMPLE_EXECUTED_TOTAL`, `MATHEMATICS::PYTHON_EVIDENCE_EXECUTED_TOTAL`, `PHYSICS::FINE_STRUCTURE_CONSTANT_RESIDUAL`, `PHYSICS::HYDROGEN_BALMER_LINE_RESIDUAL`, `PHYSICS::TRANSPORT_SCALING_RESIDUAL`.

Numerical instantiation and prediction table. The table below is a promoted subset of the numerical backing rows in Appendix **??**.

Quantity	Symbol	Value / range	Units or note
Paradigm-coherence threshold	Θ_{par}	0.15–0.25	dimensionless
Anomaly accumulation rate	J_{anom}	10^{-4}–10^{-2}	per cycle
Scientific-memory half-life	τ_{mem}	30–200	years
Model-validation bandwidth	B_{val}	0.1–1	normalized

Falsifier and collapse boundary. K9 collapses if anomaly load outruns lawful model revision or if scientific memory and validation bandwidth fail to stabilize the knowledge packet. Falsifier refs: `APPENDIX_Q_SYNTHESIS_DOSSIER_K9`, `content/falsifiability/falsifiability_master.tex`, `content/falsifiability/falsifiability_k0.tex`, `content/falsifiability/falsifiability_k9.tex`, `content/falsifiability/falsifiability_k10.tex`, `content/falsifiability/falsifiability_k11.tex`, `content/falsifiability/falsifiability_k12.tex`, `content/falsifiability/falsifiability_k1.tex`, `content/falsifiability/falsifiability_k2.tex`.

Closed-domain bindings. `MATHEMATICS`: class `THEOREM_NATIVE`, verdict `PASS`, held-out cases `0`, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/mathematics`. Metrics: `strict_campaign_acceptance_status`=PASS_31_OF_31_TERMINALIZED, `strict_terminalized_total`=31. `PHYSICS`: class `THEOREM_NATIVE`, verdict `PASS`, held-out cases `20`, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/physics`. Metrics: `covered_case_total`=30, `covered_held_out_case_total`=20, `normalized_error_mean_abs`=0.188136, `normalized_error_p95`=0.64778, `normalized_error_max`=1.48, `tail_breach_count`=0.

C.11 Level K10: formal systems and recursion

Current closure status: `PASS`. Promotion blockers: none.

Theorem claim. K10 binds bounded recursion, consistency thresholds, and proof-flow capacity into the formal-system lane of the unified-science closure stack.

Operator and K-level binding. F, Q, and U bind recursion depth, proof throughput, and consistency thresholds in formal continua. Kernel refs: `content/03_model.tex`, `content/10_klevels_full.tex`, `content/11_operators_full.tex`, `APPENDIX_Q_SYNTHESIS_DOSSIER_K10`.

Parameter law.

$$d_{\text{rec}} \in [10, 10^4], \quad \Theta_{\text{cons}} \in [0.01, 0.05], \quad J_{\text{proof}} \in [1, 10^3]$$

Observable binding and units. Recursion depth, consistency threshold, and proof-flow capacity in bounded formal systems. Observable ids: `K10::RECURSION_DEPTH`, `K10::CONSISTENCY_THRESHOLD`, `K10::PROOF_FLOW_CAPACITY`, `MATHEMATICS::EPSILON_STAR`, `MATHEMATICS::MINIMAL_DENOMINATOR`, `MATHEMATICS::MINIMAL_AUGMENTATION_RANK`, `MATHEMATICS::COMPONENT_FIBER_COUNT`, `MATHEMATICS::MOVE_CONNECTIVITY_OR_DEGREE_BOUND`, `MATHEMATICS::STRICT_TERMINALIZED_TOTAL`, `MATHEMATICS::COUNTEREXAMPLE_EXECUTED_TOTAL`, `MATHEMATICS::PYTHON_EVIDENCE_EXECUTED_TOTAL`, `PHYSICS::FINE_STRUCTURE_CONSTANT_RESIDUAL`, `PHYSICS::HYDROGEN_BALMER_LINE_RESIDUAL`, `PHYSICS::TRANSPORT_SCALING_RESIDUAL`.

Numerical instantiation and prediction table. The table below is a promoted subset of the numerical backing rows in Appendix **??**.

Quantity	Symbol	Value / range	Units or note
Recursion depth (effective)	d_{rec}	10–10^4	steps
Consistency threshold	Θ_{cons}	0.01–0.05	dimensionless
Proof-flow capacity	J_{proof}	1–10^3	steps/s

Falsifier and collapse boundary. K10 collapses if formal recursion and proof throughput cannot remain bounded without violating consistency thresholds. Falsifier refs: APPENDIX_Q_SYNTHESIS_DOSSIER_K10, content/falsifiability/falsifiability_master.tex, content/falsifiability/falsifiability_k0.tex, content/falsifiability/falsifiability_k9.tex, content/falsifiability/falsifiability_k10.tex, content/falsifiability/falsifiability_k11.tex, content/falsifiability/falsifiability_k12.tex, content/falsifiability/falsifiability_k1.tex, content/falsifiability/falsifiability_k2.tex.

Closed-domain bindings. MATHEMATICS: class THEOREM_NATIVE, verdict PASS, held-out cases 0, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/mathematics. Metrics: strict_campaign_acceptance_status=PASS_31_OF_31_TERMINALIZED, strict_terminalized_total=31. METAONTOLOGY: class FRAME_ONLY, verdict PASS, held-out cases 0, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/metaontology. Metrics: none. PHYSICS: class THEOREM_NATIVE, verdict PASS, held-out cases 20, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/physics. Metrics: covered_case_total=30, covered_held_out_case_total=20, normalized_error_mean_abs=0.188136, normalized_error_p95=0.64778, normalized_error_max=1.48, tail_breach_count=0.

C.12 Level K11: meta-theoretical reflexivity

Current closure status: PASS. Promotion blockers: none.

Theorem claim. K11 captures meta-theoretical reflexivity as irreducible scientific work beyond K10 across the closed domain atlas.

Operator and K-level binding. Q, R, and S bind meta-coherence, reflexive depth, and cross-landscape coupling under the requirement that no K10-only translation preserves all declared upper-level burdens. Kernel refs: content/03_model.tex, content/10_klevels_full.tex, content/11_operators_full.tex, APPENDIX_Q_SYNTHESIS_DOSSIER_K11.

Parameter law.

$$\Theta_{\text{meta}} \in [0.02, 0.1], \quad d_{\text{refl}} \in [3, 50], \quad J_X \in [0.1, 1]$$
$$P_{\text{funct}} \in [10^0, 10^3]$$

Observable binding and units. Meta-coherence, reflexive depth, and cross-landscape coupling in the meta-theoretical packet. Observable ids: K11::META_COHERENCE_THRESHOLD, K11::REFLEXIVE_DEPTH, K11::CROSS_LANDSCAPE_COUPLING, K11::FUNCTORIAL_POTENTIAL, MATHEMATICS::EPSILON_STAR, MATHEMATICS::MINIMAL_DENOMINATOR, MATHEMATICS::MINIMAL_AUGMENTATION_RANK, MATHEMATICS::COMPONENT_FIBER_COUNT, MATHEMATICS::MOVE_CONNECTIVITY_OR_DEGREE_BOUND, MATHEMATICS::STRICT_TERMINALIZED_TOTAL, MATHEMATICS::COUNTEREXAMPLE_EXECUTED_TOTAL, MATHEMATICS::PYTHON_EVIDENCE_EXECUTED_TOTAL, PHYSICS::FINE_STRUCTURE_CONSTANT_RESIDUAL, PHYSICS::HYDROGEN_BALMER_LINE_RESIDUAL, PHYSICS::TRANSPORT_SCALING_RESIDUAL, CHEMISTRY::GAS_PHASE_ENTHALPY_RESIDUAL, CHEMISTRY::SPECTRAL_TRANSITION_RESIDUAL, CHEMISTRY::KINETIC_OR_EQUILIBRIUM_RESIDUAL, BIOLOGY::EXPRESSION_SIGNATURE_RESIDUAL, BIOLOGY::STATE_TRANSITION_ORDER_ERROR, BIOLOGY::RESPONSE_ONSET_RESIDUAL, SYSTEMS_CIVILIZATIONAL_PROJECTION::POPULATION_TRAJECTORY_RESIDUAL, SYSTEMS_CIVILIZATIONAL_PROJECTION::GDP_GROWTH_TURNING_POINT_ERROR, SYSTEMS_CIVILIZATIONAL_PROJECTION::PROCESS_SYSTEM_REGIME_SCORE.

Numerical instantiation and prediction table. The table below is a promoted subset of the numerical backing rows in Appendix **??**.

Quantity	Symbol	Value / range	Units or note
Meta-coherence threshold	Θ_{meta}	0.02–0.1	dimensionless
Functorial potential	P_{funct}	10^0–10^3	abstract units
Reflexive depth	d_{refl}	3–50	levels
Cross-landscape coupling	J_X	0.1–1	dimensionless

Falsifier and collapse boundary. K11 collapses if an explicit K10-only reduction preserves the declared meta-theoretical burdens or if meta-coherence depends on an excluded lift route. Falsifier refs: `APPENDIX_Q_SYNTHESIS_DOSSIER_K11`, `content/falsifiability/falsifiability_master.tex`, `content/falsifiability/falsifiability_k0.tex`, `content/falsifiability/falsifiability_k9.tex`, `content/falsifiability/falsifiability_k10.tex`, `content/falsifiability/falsifiability_k11.tex`, `content/falsifiability/falsifiability_k12.tex`, `content/falsifiability/falsifiability_k1.tex`, `content/falsifiability/falsifiability_k2.tex`, `content/falsifiability/falsifiability_k3.tex`, `content/falsifiability/falsifiability_k4.tex`, `content/falsifiability/falsifiability_k5.tex`, `content/falsifiability/falsifiability_k6.tex`, `content/falsifiability/falsifiability_k7.tex`, `content/falsifiability/falsifiability_k8.tex`.

Closed-domain bindings. `MATHEMATICS`: class `THEOREM_NATIVE`, verdict `PASS`, held-out cases `0`, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/mathematics`. Metrics: `strict_campaign_acceptance_status`=PASS_31_OF_31_TERMINALIZED, `strict_terminalized_total`=31. `PHYSICS`: class `THEOREM_NATIVE`, verdict `PASS`, held-out cases `20`, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/physics`. Metrics: `covered_case_total`=30, `covered_held_out_case_total`=20, `normalized_error_mean_abs`=0.188136, `normalized_error_p95`=0.64778, `normalized_error_max`=1.48, `tail_breach_count`=0. `CHEMISTRY`: class `THEOREM_NATIVE`, verdict `PASS`, held-out cases `19`, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/chemistry`. Metrics: `covered_case_total`=30, `covered_held_out_case_total`=19, `normalized_error_mean_abs`=0.533285, `normalized_error_p95`=1.0, `normalized_error_max`=1.597772, `tail_breach_count`=0. `BIOLOGY`: class `THEOREM_NATIVE`, verdict `PASS`, held-out cases `16`, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/biology`. Metrics: `covered_case_total`=30, `covered_held_out_case_total`=16, `normalized_error_mean_abs`=0.444222, `normalized_error_p95`=1.030033, `normalized_error_max`=1.438582, `tail_breach_count`=0. `SYSTEMS_CIVILIZATIONAL_PROJECTION`: class `THEOREM_NATIVE`, verdict `PASS`, held-out cases `18`, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/systems_civilizational_projection`. Metrics: `covered_case_total`=30, `covered_held_out_case_total`=18, `normalized_error_mean_abs`=0.56097, `normalized_error_p95`=1.924704, `normalized_error_max`=2.25395, `tail_breach_count`=0. `METAONTOLOGY`: class `FRAME_ONLY`, verdict `PASS`, held-out cases `0`, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/metaontology`. Metrics: none.

C.13 Level K12: global semantic coherence

Current closure status: `PASS`. Promotion blockers: none.

Theorem claim. K12 packages global semantic coherence across the closed domain atlas because cross-domain meaning remains jointly interpretable and no surviving global falsifier breaks the unified synthesis.

Operator and K-level binding. Q, R, S, and U bind semantic coherence, global compatibility, and reference stability into the final unified-science synthesis layer. Kernel refs: `content/03_model.tex`, `content/10_klevels_full.tex`, `content/11_operators_full.tex`, `APPENDIX_Q_SYNTHESIS_DOSSIER_K12`.

Parameter law.

$$C_{\text{uni}} \in [10^3, 10^{12}], \quad \Theta_{\text{uni}} \in [10^{-3}, 10^{-2}], \quad R_{\text{uni}} \in [0.1, 1]$$
$$T_{\text{uni}} \in [10^2, 10^8]$$

Observable binding and units. Universal integration capacity, cross-continuum compatibility, and structural reachability. Observable ids: `K12::UNIVERSAL_INTEGRATION_CAPACITY`, `K12::CROSS_CONTINUUM_COMPATIBILITY`, `K12::STRUCTURAL_REACHABILITY`, `K12::GLOBAL_TENSION_BUDGET`, `MATHEMATICS::EPSILON_STAR`, `MATHEMATICS::MINIMAL_DENOMINATOR`, `MATHEMATICS::MINIMAL_AUGMENTATION_RANK`, `MATHEMATICS::COMPONENT_FIBER_COUNT`, `MATHEMATICS::MOVE_CONNECTIVITY_OR_DEGREE_BOUND`, `MATHEMATICS::STRICT_TERMINALIZED_TOTAL`, `MATHEMATICS::COUNTEREXAMPLE_EXECUTED_TOTAL`, `MATHEMATICS::PYTHON_EVIDENCE_EXECUTED_TOTAL`, `PHYSICS::FINE_STRUCTURE_CONSTANT_RESIDUAL`, `PHYSICS::HYDROGEN_BALMER_LINE_RESIDUAL`, `PHYSICS::TRANSPORT_SCALING_RESIDUAL`, `CHEMISTRY::GAS_PHASE_ENTHALPY_RESIDUAL`, `CHEMISTRY::SPECTRAL_TRANSITION_RESIDUAL`, `CHEMISTRY::KINETIC_OR_EQUILIBRIUM_RESIDUAL`, `BIOLOGY::EXPRESSION_SIGNATURE_RESIDUAL`, `BIOLOGY::STATE_TRANSITION_ORDER_ERROR`, `BIOLOGY::RESPONSE_ONSET_RESIDUAL`, `SYSTEMS_CIVILIZATIONAL_PROJECTION::POPULATION_TRAJECTORY_RESIDUAL`, `SYSTEMS_CIVILIZATIONAL_PROJECTION::GDP_GROWTH_TURNING_POINT_ERROR`, `SYSTEMS_CIVILIZATIONAL_PROJECTION::PROCESS_SYSTEM_REGIME_SCORE`.

Numerical instantiation and prediction table. The table below is a promoted subset of the numerical backing rows in Appendix **??**.

Quantity	Symbol	Value / range	Units or note
Universal integration capacity	C_{uni}	10^3–10^{12}	dimensionless
Cross-continuum compatibility	Θ_{uni}	10^{-3}–10^{-2}	dimensionless
Structural reachability	R_{uni}	0.1–1	dimensionless
Global tension budget	T_{uni}	10^2–10^8	dimensionless

Falsifier and collapse boundary. K12 collapses if cross-domain compatibility fragments, if the unified atlas cannot pass integrability, or if a surviving global falsifier breaks semantic coherence. Falsifier refs: `APPENDIX_Q_SYNTHESIS_DOSSIER_K12`, `content/falsifiability/falsifiability_master.tex`, `content/falsifiability/falsifiability_k0.tex`, `content/falsifiability/falsifiability_k9.tex`, `content/falsifiability/falsifiability_k10.tex`, `content/falsifiability/falsifiability_k11.tex`, `content/falsifiability/falsifiability_k12.tex`, `content/falsifiability/falsifiability_k1.tex`, `content/falsifiability/falsifiability_k2.tex`, `content/falsifiability/falsifiability_k3.tex`, `content/falsifiability/falsifiability_k4.tex`, `content/falsifiability/falsifiability_k5.t` ex, content/falsifiability/falsifiability_k6.tex, content/falsifiability/falsifiability_k7.tex, content/falsifiability/falsifiability_k8.tex.

Closed-domain bindings. MATHEMATICS: class THEOREM_NATIVE, verdict PASS, held-out cases 0, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/mathematics. Metrics: strict_campaign_acceptance_status=PASS_31_OF_31_TERMINALIZED, strict_terminalized_total=31. PHYSICS: class THEOREM_NATIVE, verdict PASS, held-out cases 20, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/physics. Metrics: covered_case_total=30, covered_held_out_case_total=20, normalized_error_mean_abs=0.188136, normalized_error_p95=0.64778, normalized_error_max=1.48, tail_breach_count=0. CHEMISTRY: class THEOREM_NATIVE, verdict PASS, held-out cases 19, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/chemistry. Metrics: covered_case_total=30, covered_held_out_case_total=19, normalized_error_mean_abs=0.533285, normalized_error_p95=1.0, normalized_error_max=1.597772, tail_breach_count=0. BIOLOGY: class THEOREM_NATIVE, verdict PASS, held-out cases 16, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/biology. Metrics: covered_case_total=30, covered_held_out_case_total=16, normalized_error_mean_abs=0.444222, normalized_error_p95=1.030033, normalized_error_max=1.438582, tail_breach_count=0. SYSTEMS_CIVILIZATIONAL_PROJECTION: class THEOREM_NATIVE, verdict PASS, held-out cases 18, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/systems_civilizational_projection. Metrics: covered_case_total=30, covered_held_out_case_total=18, normalized_error_mean_abs=0.56097, normalized_error_p95=1.924704, normalized_error_max=2.25395, tail_breach_count=0. METAONTOLOGY: class FRAME_ONLY, verdict PASS, held-out cases 0, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/metaontology. Metrics: none.

D OC 1.3.1 Critique Closure Appendix

- critique_id=CRITIQUE::DESKTOP_CLOSEOUT_STATE_DRIFT; processing_class=REAL_PACKAGING_WEAK
  output_mode=clarification_unit; publication_effect=none
- critique_id=CRITIQUE::PUBLIC_REPO_ARCHITECTURE_DRIFT; processing_class=REAL_PACKAGING_WEA
  output_mode=clarification_unit; publication_effect=refresh public architecture truth and release gate explanation
- critique_id=CRITIQUE::SCOPE_INFLATION_CHECK__WEDGE_001; processing_class=REQUIRES_DEMOTIO
  output_mode=demotion_decision; publication_effect=bounded clarification note for reviewer/public audience
- critique_id=CRITIQUE::PUBLIC_REPO_SIMULATION_ABSENCE; processing_class=REAL_PACKAGING_WE
  output_mode=clarification_unit; publication_effect=bind simulation corpus into OC Core 1.3.1 release package

E OC 1.3.1 Science Delta Appendix

- delta_id=DELTA::WHITE_SPOT_CLOSURE; delta_class=scientific_closure; status_1_3_1=closure_companion_and
  action_bucket=must_be_added

- delta_id=DELTA::SIMULATION_CORPUS; delta_class=simulation_and_reproducibility; status_1_3_1=top_level_simulation_corpus; action_bucket=must_be_added
- delta_id=DELTA::DATA_MANIFEST_LAYER; delta_class=data_route_and_traceability; status_1_3_1=manifest_anc action_bucket=must_be_added
- delta_id=DELTA::MASTER_MONOGRAPH; delta_class=release_authority; status_1_3_1=oc_core_1_3_1_master_p action_bucket=must_be_added
- delta_id=DELTA::ARTICLE_FAMILY; delta_class=publication_surface; status_1_3_1=bounded_article_family_prese action_bucket=must_be_added
- delta_id=DELTA::RELEASE_AUTOMATION; delta_class=release_automation; status_1_3_1=contract_and_validato action_bucket=must_be_added
- delta_id=DELTA::NO_SEND_OUTBOUND; delta_class=outbound_readiness; status_1_3_1=release_specific_packag action_bucket=must_be_added
- delta_id=DELTA::PUBLIC_RELEASE_STAGE; delta_class=release_staging; status_1_3_1=oc_core_1_3_1_ready_n action_bucket=candidate_for_1_3_1
- delta_id=DELTA::K11_K12_EXTENSION; delta_class=high_cost_theory_extension; status_1_3_1=mapped_but_not action_bucket=must_remain_frontier
- delta_id=DELTA::PAID_DATA_ESCALATIONS; delta_class=data_escalation; status_1_3_1=still_optional; action_bucket=too_expensive_now